



Decision Support

Soft consensus cost models for group decision making and economic interpretations

Huanhuan Zhang^a, Gang Kou^b, Yi Peng^{a,*}^a School of Management and Economics, University of Electronic Science and Technology of China, Chengdu 610054, China^b School of Business Administration, Southwestern University of Finance and Economics, Chengdu 611130, China

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ABSTRACT

In a group decision-making (GDM) process, experts reach a consensus after discussion and persuasion, which requires a moderator to spend time and resource to persuade experts to change their original opinions. Since a unanimous agreement is hard to achieve and costly, a consensus degree or soft consensus was used and various approaches have been proposed to measure the level of consensus in GDM. Though cost is an important factor in GDM, few works have calculated consensus cost occurred during the process. Moreover, the degree of consensus was not considered in the minimum cost consensus study. The objective of this paper is to propose consensus models under a certain degree of consensus, which considers both consensus degree and cost in GDM. To do this, we develop a generalized soft cost consensus model under a certain degree of consensus, which is built by defining a consensus level function and a generalized aggregation operator. A soft minimum cost consensus model is constructed based on arithmetic weighted average operator (AWAO), and the maximum return model is constructed through its dual model. The cost (compensation) is studied from both the perspectives of a moderator and individual experts. The relationships between the two soft consensus cost models are analyzed, and the economic significance of the models are also discussed. Numerical examples are used to explain the proposed models. In addition, to show the usability of the proposed models in real-world context, we apply the proposed models to a loan consensus scenario using data from an online peer-to-peer lending platform.

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1. Introduction

Group decision-making (GDM) aims at solving unstructured decision-making problems that require the participation of experts or decision makers. A decision-making group needs to take a broad and thorough discussion of the issues before reaching a consensus. Generally, there are two processes in GDM: consensus process and selection process. Experts try to achieve the greatest degree of agreement on the issues in a consensus process, in which experts express, discuss, and modify their preferences for several rounds (Ben-Arieh, Easton, & Evans, 2009; Cabrerizo, Moreno, Pérez, & Herrera-Viedma, 2010; Dong & Cooper, 2016; Dong, Zhang, & Herrera-Viedma, 2016a; Herrera & Herrera-Viedma, 1996; Herrera, Herrera-Viedma, & Verdegay, 1997a; Herrera-Viedma, Herrera, & Chiclana, 2002; Pérez, Cabrerizo, Alonso, & Herrera-Viedma,

2014; Tapia García, Del Moral, Martínez, & Herrera-Viedma, 2012; Wu & Kou, 2016). A consensus process is normally guided by a moderator, who does not participate in discussions but knows the agreement in each moment of a consensus process. A moderator oversees and guides a consensus process to achieve the maximum possible agreement and reduce the number of experts outside the consensus (Cabrerizo et al., 2010; Dong & Cooper, 2016; Herrera & Herrera-Viedma, 1996; Herrera, Herrera-Viedma, & Verdegay, 1996; Kacprzyk, Fedrizzi, & Nurmi, 1992; Xu, Cabrerizo, & Herrera-Viedma, 2017). The selection process obtains a set of solutions from experts' opinions, which involves aggregation of individual opinions and exploitation of a collective opinion (Cabrerizo et al., 2017; Cabrerizo et al., 2010; Cheng, Chen, & Chiang, 2016; Herrera, Herrera-Viedma, & Verdegay, 1998; Herrera-Viedma, Cabrerizo, Kacprzyk, & Pedrycz, 2014; Roubens, 1997). The aggregation of individual information can be classified into two groups: aggregation of preferences, which involves a group of decision makers (DMs) (Cabrerizo et al., 2017; Dong, Zhang, Hong, & Xu, 2010; Dong et al., 2016a; Ma, 2016; Zhou, Wang, Li, & Wang, 2016); and aggregation

* Corresponding author.

E-mail addresses: kougang@swufe.edu.cn (G. Kou), pengyi@uestc.edu.cn (Y. Peng).

of knowledge (opinion), which involves some experts (Ben-Arieh & Chen, 2006; Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009; Chiclana, Herrera-Viedma, Herrera, & Alonso, 2007; Dong, Zhang, & Herrera-Viedma, 2016b; Zhang, Dong, & Xu, 2013; Zhang, Dong, Xu, & Li, 2011). This article mainly focuses on the aggregation of the experts' opinions, which are expressed in numerical numbers.

Traditionally, consensus is defined as a full and unanimous agreement (Bezdek, Spillman, & Spillman, 1978; Kacprzyk & Fedrizzi, 1989; Spillman, Spillman, & Bezdek, 1980). Unfortunately, this type of consensus is very difficult to achieve. Even if an ideal agreement can be achieved, the consensus reaching process could be too long and unacceptably expensive (Bordogna, Fedrizzi, & Pasi, 1997; Herrera et al., 1997a; Kacprzyk & Fedrizzi, 1989; Tapia García et al., 2012). In practice, a consensus degree is used to measure the level of consensus in GDM and various approaches have been proposed. The first type of approach is hard consensus measures. Several authors introduced hard consensus measures varying from 0 to 1 (no consensus and full consensus) (Bezdek et al., 1978; 1979; Spillman, Bezdek, & Spillman, 1979; Spillman et al., 1980). In Bezdek et al. (1978, 1979), the authors defined scalars to measure individual preferences and group consensus. In Spillman et al. (1979), the authors proposed some measures of attitudinal similarities between individuals that is an extension of the classical Tanimoto coefficient, and utilized fuzzy preference matrices and α -level sets to determine coalitions over time. In Spillman et al. (1980), the authors developed a consensus measure based on α -cuts of the respective individual fuzzy preference matrices and construction of α -Consensus measures from fuzzy relations.

The second type is soft consensus measures. Early research was based on the concept of coincidence to measure consensus (Kacprzyk, 1987) and reflected a large spectrum of partial agreements in GDM. Different coincidences were considered in the literature. Some based on strict coincidence, that is, accepting only total coincidence or null coincidence (Fedrizzi, Kacprzyk, & Nurmi, 1993; Herrera & Herrera-Viedma, 1996; Herrera, Herrera-Viedma, & Verdegay, 1997b; Kacprzyk, 1987). Others based on less strict coincidence, that is, accepting different coincidence degrees (Fedrizzi et al., 1993; Herrera et al., 1997a; Kacprzyk, 1987). In soft consensus measures, experts' evaluations of alternatives are specified as individual preference relations and the degree of consensus indicates similarities between preference relations (Bordogna et al., 1997; Cabrerizo et al., 2017; Chao, Kou, Li, & Peng, 2018; Fedrizzi & Kacprzyk, 1988; Kacprzyk & Fedrizzi, 1989; Kou, Ergu, & Shang, 2014a; Kou & Lin, 2014; Yager, 1983; Zadeh, 1983). When experts' opinions are expressed as preference relations in a GDM problem, individual consistency and group consensus are widely used to measure the degree of consensus that has been reached in a GDM situation.

In recent years, notable progress has been made in the consensus level measures in preference relations. Sun and Ma (2015) provided a way of evaluating the consensus degree in multi-attribute GDM with linguistic information. Wu and Xu (2016) defined a hesitant fuzzy linguistic preference relation (HFLPR) consistency measure, and developed a convergent consistency improving process to deal with HFLPR individual rationality and group rationality. Zhou et al. (2016) defined an intuitionistic hesitant linguistic number (IHLN) that contains a linguistic term, a set of membership degrees and a set of non-membership degrees. It can express the consensus and disagreements of experts in a flexible way. Wu and Xu (2012) introduced the concept of an individual consistency index and a group consensus index to measure the consensus level, and designed a decision support model to deal with the group consensus process. Dong et al. (2010) defined consensus indexes to measure consensus degree among experts for the AHP group decision making using row geometric mean prioritization method. Xu and Yager (2009) developed a similarity

measure between intuitionistic fuzzy sets for consensus analysis based on intuitionistic fuzzy preference relations, which extended the similarity measure to interval-valued intuitionistic fuzzy situations to assess the consensus of experts' preferences. In Cabrerizo, Alonso, and Herrera-Viedma (2009), consensus degrees were measured at three different levels: pairs of alternatives, alternatives, and relations. They were used to support experts when they want to change their opinions, and evaluate the agreement between individual expert's opinion and group opinion. In Kou, Lu, Peng, and Shi (2012) and Kou, Peng, and Wang (2014), the authors considered algorithm selection problem as a multiple criteria decision making (MCDM) problem and presented MCDM-based approach to rank a selection of different algorithms or methods.

Consensus process involves dynamic opinion exchange and communication between experts and a moderator (Gong, Xu, Li, & Xu, 2015). Under the advice and guidance of the moderator, experts change their opinions step by step until certain consensus is reached (Cabrerizo et al., 2010; Herrera-Viedma et al., 2014). Since the consensus process is complex and resource consuming, cost is an important issue in GDM (Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009; Kou, Ergu, Lin, and Chen (2016)). From a moderator's point of view, he/she takes effective measures, such as economic compensations to experts (also referred to as 'consensus cost'), to make experts gradually change their opinions and finally reach a consensus. From experts' points of view, they sacrifice personal interests and contribute to the collective consensus, thus deserve to be compensated or rewarded (Gong, Zhang, Forrest, Li, & Xu, 2015). From the perspective of resource consumption, a moderator seeks to keep the total cost (e.g., time and fees) as low as possible in order to reach a consensus (Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009; Liu, Chan, Li, Zhang, & Deng, 2012; Zhang et al., 2011), while individual expert hopes to maximize his/her interest or return for the opinion change. To capture the cost and compensation in the consensus process and allow a certain degree of consensus in GDM, this paper constructs two kinds of soft cost consensus models from the perspective of a moderator and experts, respectively. The relationships between the two models are also studied.

In many real-life GDM problems, acceptable agreements achieved on minimum cost consensus are more effective and desirable than more time-consuming and expensive unanimous agreements. Therefore, both degree of consensus and cost of consensus are important factors in GDM. However, the study of consensus degrees provided no calculation of consensus cost, while the minimum cost consensus study in the literature did not consider degree of consensus (Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009; Gong, Zhang et al., 2015). To the best of our knowledge, no work has considered both factors. The soft cost consensus models extend Ben-Arieh's minimum cost consensus theory (Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009) and Gong's consensus model (Gong, Zhang et al., 2015) to a certain consensus level.

The proposed models in this paper have the following advantages: (1) Soft consensus measure is adopted by defining the consensus level function instead of opinion deviation, which intuitively reflect the degree of consensus. Since an expert's opinion only needs to reach the consensus level expected by a moderator and the consensus process can be ended, it is more realistic than the complete consensus cost models, and expresses the consensus of experts in a more flexible way. (2) The models are constructed from both moderators and experts' points of view. Thus, the cost (compensation) can be explored both from the viewpoints of moderators and experts, and the relationships between the cost for a consensus paid by a moderator and compensations for experts can also be studied. (3) The models show the consumption of resources and the changes of experts' opinions aggregation under different levels of consensus both in the linear and non-linear consensus environment, which is useful for moderators to optimize the resource

allocations. (4) Aggregation operators are introduced to obtain the ideal group consensus opinion and reflect the importance of different experts' opinions through different weights. Aggregation operators can also reveal the relationships between individuals' opinions and the group consensus opinion in some decision problems.

Minimum cost consensus has numerous applications in real-world situations. Researchers have conducted some studies using minimum cost consensus for road junction control method selection (Kwok & Lau, 2016), relocation negotiations of urban housing demolition (Gong, Xu, Chiclana, & Xu, 2017), pollutant reduction negotiations between the government and manufacturing enterprises (Gong, Xu, Li, et al., 2015), and trans-boundary pollution control negotiations (Cheng, Zhou, Cheng, Zhou, & Xie, 2018). In addition to the applications mentioned above, minimum cost consensus can be applied in many other areas.

To illustrate the usefulness of the proposed models, this study designs a numerical example to interpret and validate the proposed models, and applies the models to a loan consensus problem in online peer-to-peer (P2P) lending. P2P lending allows individuals to directly lend to and borrow from each other on an internet-based platform. A key challenge is how the supply and demand of funds can be effectively matched for lenders and borrowers with appropriate interest rates (Wei & Lin, 2016), which is called the *loan consensus problem* in this paper. Normally, lenders lend their money to multiple borrowers with different credit risks and returns, and they use an interval range to express their desired interest rates. Borrowers, on the other hand, apply for loans with fixed interest rates. P2P lending platforms serve as moderators, who adopt compensation strategies to coordinate both sides of loans (lenders and borrowers) and lead them to reach consensus. The features of loan consensus in P2P lending are much in line with the soft minimum cost consensus problem. The proposed soft consensus models are used to help match the supply and demand of funds in P2P lending effectively using real-world data from a Chinese P2P lending platform.

The rest of this paper is organized as follows. Section 2 introduces related works and the principles and basic assumptions of the minimum cost consensus model. Section 3 describes the construction of a generalized soft minimum cost consensus model under a certain degree of consensus, and builds three soft minimum cost models to show the changes of the consensus costs and experts' opinions aggregation with numerical examples. The properties and economic significance of the soft minimum cost consensus model under a certain degree of consensus are also explored. Section 4 develops the maximum return model under a certain level of consensus and provides its economic interpretation. Furthermore, the relationships between the soft minimum cost and maximum return models under a certain level of consensus are also explored using a numerical example. Section 5 applies the proposed soft consensus models to a real-world loan consensus of online P2P lending problem to assist the lenders and borrowers in reaching a certain level of consensus on interest rates. Section 6 concludes the paper.

2. Related works and preliminaries

This section introduces the concept of minimum cost consensus, and existing works on minimum cost models and maximum return models based on the linear programming.

2.1. The minimum cost consensus

In 2007, Ben-Arieh and Easton (2007) and Ben-Arieh et al. (2009) introduced the concept of minimum cost consensus, which describes a linear cost of changing experts' opinions considering a single and multi-criteria decision consensus. Suppose that there

are m experts $E = \{e_1, \dots, e_m\}$ in a GDM problem. Let $o_i \in R$ represents the opinion of expert e_i ($i \in M = \{1, 2, \dots, m\}$) and o' represents the consensus opinion. It assumes that an expert's opinion can be captured numerically as a quantitative value in the decision process. An expert's opinion can be measured and modified by the guidance of a moderator, who knows the values of experts' opinions and tries to direct a consensus process to achieve the maximum possible agreement. Without loss of generality, we assume that the opinions are ordered on the number line; thus, $o_1 \leq o_2 \leq \dots \leq o_m$. A consensus is defined as when all experts have the same opinion, $o'_1 = o'_2 = \dots = o'_m = o'$. For each expert, c_i is a unit cost paid by a moderator to persuade expert e_i ($i \in M$) to change his/her opinion. Based on the distance between experts' opinions and group consensus, the MCC problem is defined as: $f_c(o') = \sum_{i=1}^m c_i |o' - o_i|$. The function $f_c(o')$ denotes the total cost paid to the experts by the moderator to arrive at a consensus. The MCC problem seeks an o^* such that the cost to reach the consensus is minimal. If the function is assumed to be quadratic and can be formulated as $\sum_{i=1}^m c_i (o' - o_i)^2$, a consensus at minimum quadratic cost (CMQC) problem can also be defined (Ben-Arieh et al., 2009).

Based on the MCC problem, Ben-Arieh and Easton (2007) and Ben-Arieh et al. (2009) defined an ε -consensus at minimum cost problem, which allows each expert's opinion to be sufficiently close to a consensus opinion. Let ε denotes the maximum units of deviations for reaching a common consensus. The minimum cost consensus model with the ε threshold constraint of opinion deviation is (Ben-Arieh and Easton, 2007):

$$\begin{aligned} \min f_{c_\varepsilon}(o') &= \sum_{i: o_i < o' - \varepsilon} c_i (o' - \varepsilon - o_i) + \sum_{i: o_i > o' + \varepsilon} c_i (o_i - o' - \varepsilon) \\ \text{s.t. } |o' - o'_i| &\leq \varepsilon \end{aligned} \quad (1)$$

The objective function $f_{c_\varepsilon}(o')$ denotes the total cost paid to experts by a moderator to arrive at a consensus within ε distance. The ε -consensus at minimum cost problem seeks an o^* such that the cost to reach this ε -consensus is minimal. This is a soft consensus because experts' opinions and consensus opinion are within certain distance deviations ε . Although this approach is useful, they cannot reflect the cost of soft consensus problems because: (1) The objective function is complicated, which cannot reflect the cost of consensus in an intuitive form, especially the cost under soft consensus. In addition, for some complicated decision-making problems, individuals cannot make reasonable judgment and understand the distance deviation of experts' opinions from the group consensus, which is unable to intuitively reflect the degree of consensus. (2) Consensus is a complex and resource-consuming process, which requires the exchange of opinions among a moderator and experts (Gong, Xu, Li, et al., 2015; Gong, Zhang et al., 2015) and needs a balance of interests of both sides. Ben-Arieh and Easton (2007) and Ben-Arieh et al. (2009) only defined the MCC problem from the viewpoint of a moderator. The study of compensation that an expert expected for changing his/her opinion is necessary and needs to be considered from the viewpoint of the expert.

Zhang et al. (2011) and Zhang et al. (2013) incorporated aggregation operators into the consensus process, and proposed a minimum cost consensus model based on the aggregation operators. Zhang, Dong, and Xu (2014) studied the multiple attribute GDM to minimize the adjustment cost and reduced consensus costs. Liu et al. (2012) proposed an optimal consensus method with minimum cost in fuzzy GDM. Li, Zhang, and Dong (2017) proposed an interactive consensus reaching process with the minimum and uncertain cost in GDM. Cheng et al. (2018) explored the minimum cost consensus models in the asymmetric costs context based on the rectilinear distance function. Although a large number of models for minimum cost consensus have been proposed, the soft consensus models or the degree of consensus for the minimum cost are not studied. Moreover, consensus requires the exchange

of opinions among a moderator and experts (Gong, Xu, Li, et al., 2015; Gong, Zhang et al., 2015) and needs a balance of interests of both sides as mentioned above. It is important to study the compensations from the viewpoints of experts.

2.2. Minimum cost model and maximum return model based on linear programming

The process of consensus requires opinion exchange and communication between experts and a moderator, and needs a balance of interests of both sides. From the viewpoint of a moderator, he/she hopes to achieve the highest level of consensus while paying the minimum total cost (Ben-Arieh & Easton, 2007; Ben-Arieh et al., 2009; Liu et al., 2012; Zhang et al., 2011). From each expert's perspective, he/she expects to gain the maximum compensation for his/her changing opinions (Gong, Xu, Lu, Li, & Xu, 2015; Gong, Xu, Zhang, et al., 2015; Gong, Zhang et al., 2015; Zhang, Gong, & Chiclana, 2017). Mathematically, these two goals are dual to each other. Let y_i , $i \in M$, denotes the unit return that expert e_i expect to be compensated. In Gong, Xu, Lu, et al. (2015), Gong, Xu, Zhang, et al. (2015), Gong, Zhang et al. (2015), and Zhang et al. (2017), the authors proposed several group decision consensus on minimum cost models and maximum return models based on linear programming from the viewpoints of moderators and individual experts. Two consensus models were constructed as follows (Gong, Xu, Zhang, et al., 2015; Gong, Zhang et al., 2015):

(a) minimum cost consensus model.

$$P(w) : \min \phi = \sum_{i=1}^m c_i |o' - o_i|$$

$$s.t. \{o' \in O \quad (2)$$

where $O = \{o' \in R | o' > 0\}$ is the set of all possible opinions. ϕ is the total costs paid to all experts.

(b) maximum return consensus model.

$$D(w) : \max Z(Y)$$

$$= [(o_1 - o')y_1, (o_2 - o')y_2, \dots, (o_m - o')y_m]$$

$$s.t. \{y_i \in R \quad (3)$$

where the feasible set $Y = \{y_i \in R\}$ denotes the set of all possible unit returns of experts. $Z = \{(o_i - o')y_i | y_i \in R\}$ is the set of total returns for each individual.

In the minimum cost consensus model and maximum return model, let o^* be the group optimal opinion, the following properties can be obtained:

Property 1. Gong, Zhang et al. (2015) The statement $y_i = -c_i < 0$ holds when $o^* > o_i$ holds; and $y_i = c_i > 0$ holds when $o^* < o_i$ holds. This denotes $|y_i| = c_i$ holds when $o^* \neq o_i$ holds; $-c_i \leq y_i \leq c_i$ holds when $o^* = o_i$ holds.

Property 2. Gong, Zhang et al. (2015) The statement $o^* \leq o_i$ holds when $y_i = c_i$; $o^* \geq o_i$ holds when $y_i = -c_i$; and $o^* = o_i$ holds when $-c_i < y_i < c_i$.

Property 1 and 2 show that there exist some relationships between the minimum cost and maximum return models. This paper clarifies the relationships between them under a certain degree of consensus. In addition, we interpret the economic significance of the models and the change of consensus cost. The experts' opinions adjustment under different levels of consensus are also explored.

The multi-objective optimization models proposed by Gong, Xu, Lu, et al. (2015), Gong, Xu, Zhang, et al. (2015), and Gong, Zhang et al. (2015) studied the minimum cost consensus problem from the perspective of a moderator, and constructed maximum return

models from the viewpoints of experts. The consensus under a certain degree of consensus (soft consensus) was not considered in these two models, though soft consensus is more flexible and much in line in real-world GDM scenarios. More importantly, consensus requires the exchange of opinions among a moderator and experts (Gong, Xu, Li, et al., 2015; Gong, Zhang et al., 2015) and needs a balance of interests of both sides as mentioned above. Under a certain level of consensus, the expected compensations from an individual expert's perspective in soft consensus also need to be studied. Actually, the minimum cost model proposed by Gong (Gong, Xu, Lu, et al., 2015; Gong, Xu, Zhang, et al., 2015; and Gong, Zhang et al., 2015) is a special case of the model proposed in this paper. The soft consensus models proposed in this paper can be transformed into Gong's model when the consensus level $\gamma = 1$.

3. Soft MCC models and economic interpretation

This section describes the construction of generalized soft minimum cost consensus model under a certain degree of consensus in detail. Three soft minimum cost models are built based on different weighted average (WA) operators, which show the changes of consensus costs and experts' opinions aggregation using numerical examples. Moreover, the economic significance of the soft MCC model under a certain degree of consensus is also explored.

3.1. Model construction

The soft MCC models proposed in this section extend the concept of minimum cost consensus described in Section 2 to a certain degree of consensus, and improve the minimum cost consensus model.

3.1.1. Definition of consensus level function

Soft consensus measure is widely used in consensus building with preference relations in GDM. For example, in Wang and Lin (2016), the authors proposed a soft consensus model by employing ratio-based similarity measurements and induced interval-valued cross ratio ordered weighted geometric (IIVCROWG) operator. The similarity level between one preference relation and others was established with a ratio-based similarity threshold. In Ma (2016), the author developed an optimization model involving maximum consensus sequences for group ranking problems. The model included agreement threshold and disagreement threshold, which provided a flexibility way for experts to arrive at a preferred compromised solution. In Cabrerizo et al. (2017), the authors studied how to adapt the existing soft consensus measures to handle group decision-making situations in unbalanced fuzzy linguistic contexts, and analyzed the advantages and drawbacks of these approaches. A comprehensive review of soft consensus models under fuzzy environments can be found in Herrera-Viedma et al. (2014). This paper proposes soft consensus measures for consensus study by defining the consensus level function. Different from the soft consensus measures with preference relations, the consensus level function aims at reflecting the approximation degree between individual expert's opinion and the consensus of a group, which are closely related to the costs in the consensus. In addition, the soft consensus models in this paper focus on the study of consensus costs from the perspective of a moderator and compensations from the perspective of experts.

Assuming that a set of m experts is $E = \{e_1, \dots, e_m\}$, let $o_i \in R$ denotes the opinion of the i th expert and the group consensus opinion is o' . Without loss of generality, this study assumes that $o_1 \leq o_2 \leq \dots \leq o_m$ (Ben-Arieh & Easton, 2007; Ben-Arieh et al.,

2009; Gong, Zhang et al., 2015). Let o'_i denotes the adjusted opinion of expert i , $i \in M = \{1, 2, \dots, m\}$. A common distance metric is used to measure the deviation of each expert's adjusted opinion from the group consensus opinion, which is expressed in the form of distance $f_i(o'_i, o') = |o'_i - o'|$. Consensus degrees of experts and consensus degrees of group decisions can then be measured. We also use distance metric to measure the deviation of each expert's opinion from his/her adjusted opinion, which is expressed in the form of distance $f_i(o_i, o'_i) = |o_i - o'_i|$. Using distance deviations to measure group consensus has two advantages. The first is that the distance deviation conforms to human decision-making intuitions. The second is easy computation. Though distance deviation presents a nonlinear form, it can be converted into a linear form to construct a consensus measure and consensus cost functions.

Definition 1. In group decision-making, individual expert's consensus level function is defined as follows:

$$f_i(o'_i, o') = \begin{cases} 1 - \frac{|o'_i - o'|}{o'} & o'_i < 2o' \\ 0 & o'_i \geq 2o' \end{cases} \quad (4)$$

where $i \in M = \{1, 2, \dots, m\}$, $o'_i \geq 0$, $o' > 0$, and $f_i(o'_i, o') \in [0, 1]$

The consensus level function reflects the approximation degree between individual expert's opinion and the consensus of a group, and therefore provides an estimation of whether the consensus of the desired degree is reached. The rationality of the function is described below:

- When $o' \leq o'_i < 2o'$, $f_i(o'_i, o') = 2 - \frac{o'_i}{o'}$, it is easy to prove $0 \leq f_i(o'_i, o') \leq 1$. The greater the deviation of o'_i from o' , the closer the function is to 0. The smaller the deviation, the closer the function is to 1. This is consistent with the level of consensus measurement.
- When $0 \leq o'_i \leq o'$, $f_i(o'_i, o') = \frac{o'_i}{o'}$, it is easy to prove $0 \leq f_i(o'_i, o') \leq 1$. The greater the deviation of o'_i from o' , the closer the function is to 0. The smaller the deviation, the closer the function is to 1. This is consistent with the level of consensus measurement.
- When $o'_i \geq 2o'$, the deviation of o'_i from o' is so big that the consensus level is defined as 0.
- Finally, in the case of $o' \leq o'_i < 2o'$ and $0 \leq o'_i \leq o'$, if the deviations of o'_i from o' in both cases are equal, there must be consensus measure function obtains the same value. This indicates that regardless of whether individual expert raises or reduces his/her opinion, as long as the distance deviations from his/her adjusted opinion to the group consensus opinion are the same, the consensus degree must be equal, which is consistent with the consensus level in reality.

Definition 1 defines the consensus degree function for an expert, and then the consensus level of a group can be expressed. We set the level of group consensus as γ , $0 \leq \gamma \leq 1$. The closer γ approaches 1, the higher group consensus it represents. When $\gamma = 1$, the group consensus is fully achieved, that is, opinions of each expert after adjustment are equal to the ideal group consensus. The closer γ approaches 0, the lower group consensus it represents. When $\gamma = 0$, the group consensus level is the lowest. Let $\gamma_i = f_i(o'_i, o')$, $i \in M = \{1, 2, \dots, m\}$ denotes the consensus degree of the i th expert, and $\gamma = \min\{\gamma_1, \gamma_2, \dots, \gamma_m\}$ represents the consensus level of the whole group.

- (i) When $\gamma = 0$, all experts are in the initial state and do not need to adjust their opinions. The optimal opinion of expert e_i is $o_i^* = o_i$, and the minimum cost of the consensus is 0.

- (ii) When $\gamma = 1$, the minimum cost consensus model degrades into the consensus model proposed in Gong, Xu, Zhang, et al. (2015) and Gong, Zhang et al. (2015), which will not be repeated here (shown in Section 2).
- (iii) When $\gamma \in (0, 1)$, the group is not fully consistent and in a certain degree of consensus, which is the focus of this paper.

3.1.2. The generalized aggregation operator

To aggregate individual expert's preference into group preferences, aggregation operators are widely used in GDM problems Dong et al. (2016b); Fedrizzi et al. (1993); Li, Kou, and Peng (2018); Peng, Kou, Wang, and Shi (2011); Zhang et al. (2011). A traditional method of aggregating experts' preferences is WA method, which considers opinions' degree of importance. Yager (1983) developed the ordered weighted averaging (OWA) operator, which has been applied and studied by a number of researchers and a variety of extended OWA operators have been proposed Chiclana et al. (2007); Herrera et al. (1996); Wang and Lin (2016); Wu and Xu (2016); Xu et al. (2017); Zhang et al. (2013); Zhou et al. (2016). OWA operators, under certain conditions, are special case of WA operators, and can be converted to WA operators. For example, if the weight vector satisfies $w = (1/n, 1/n, \dots, 1/n)^T$, the OWA operator will degenerate into an arithmetic average operator. By using the most common and basic arithmetic weighted average operator (AWAO), geometry weighted average operator (GWAO) and harmonic weighted average operator (HWAO), three soft minimum cost consensus model are constructed as examples to show the changes of the consensus costs and experts' opinions aggregation with linear and non-linear models.

This article mainly uses the AWAO as a method of aggregating group opinions. By constructing its corresponding maximum return model, this paper focuses on the compensation mechanism in the soft cost consensus models. There are three main advantages of using AWAO to aggregate group opinions. First, the AWAO is a common aggregation operator, which is frequently used in real life and many other operators evolve from it. Second, it intuitively reflects the relationship between individual expert's opinion and group consensus opinion, which is easily understood and accepted by the public. Third, the dual form for the minimum cost consensus can be easily obtained using the AWAO. Based on the duality theory of linear programming, it can be well explored from the perspectives of experts under a certain degree of consensus. The relationships between the two soft consensus cost models (minimum cost and maximum return models) can then be analyzed, which is one of the objectives of this article. For more details about minimum-cost consensus models under aggregation operators, please refer to Zhang et al. (2011).

From the above discussion, we can define a general form of the aggregation function:

Definition 2. Let o'_i denotes the adjusted opinion of expert i , $i \in M = \{1, 2, \dots, m\}$, and w_i , $i \in M = \{1, 2, \dots, m\}$ denotes the weights of experts' opinions in the minimum cost consensus problem, which reflects the degree of importance for each expert. The aggregation operator function can be defined as $g_i = \sum_{i=1}^m w_i \otimes o'_i$, $i \in M = \{1, 2, \dots, m\}$. Where $\otimes$ denotes the general aggregation operator Ben-Arieh and Chen (2006).

3.1.3. The construction of soft MCC models

Based on the above analysis, a generalized form of soft MCC model is constructed as follows.

Definition 3. Let $o'_i, o_i, o', \gamma, w_i$ be defined as above, and c_i denotes a unit cost paid by a moderator. The generalized soft minimum

cost consensus model under a certain consensus degree γ (MCC- γ model) is defined as:

$$\begin{aligned} \min \phi &= \sum_{i=1}^m c_i |o'_i - o_i| \\ \text{s.t.} \quad &\begin{cases} 1 - \frac{|o'_i - o_i|}{o'_i} \geq \gamma & (5-1) \\ o' = \sum_{i=1}^m w_i \otimes o'_i & (5-2) \\ o'_i, o' \geq 0 \end{cases} \end{aligned} \quad (5)$$

The objective function $\phi = \sum_{i=1}^m c_i |o'_i - o_i|$ represents the minimum total cost of consensus for a group. The restriction condition (5-1) is the consensus level function (Section 3.1.1). The constraint (5-2) shows that the consensus opinion is obtained using the aggregation operator (Section 3.1.2). Under these constraints, the total cost of achieving group consensus is as small as possible. Different from the previous consensus studies of minimum cost, this paper considers that a group consensus is accomplished when the deviation between each expert's opinion and the ideal group consensus is relatively small, which is reflected by the consensus level function. In addition, the relationship between experts' opinions and a consensus opinion under a certain level of consensus is built using aggregation operators.

Using different WA operators to assemble the group consensus optimal opinion, we construct the following three group consensus models minimizing total cost under a certain level of consensus. A general property will be given in [Theorem 1](#). Moreover, we will compare resource consumptions and the changes of experts' opinions aggregation under the three models in [Section 3.3](#).

- (a) The first model uses arithmetic average operator to assemble the consensus opinion of a group. Let $o_i^a, i \in M$ be the adjusted opinion of the i th expert and o^a be the group consensus opinion. The soft consensus model of minimum cost based on AWAO under the γ level (the MCC-AWAO- γ model) is defined as:

$$\begin{aligned} \min \phi &= \sum_{i=1}^m c_i |o_i^a - o_i| \\ \text{s.t.} \quad &\begin{cases} 1 - \frac{|o_i^a - o_i|}{o_i^a} \geq \gamma & (6-1) \\ o^a = \sum_{i=1}^m w_i o_i^a & (6-2) \\ o_i^a, o^a \geq 0 \end{cases} \end{aligned} \quad (6)$$

- (b) The second model uses geometric average operator to assemble the consensus opinion of a group. Let $o_i^g, i \in M$ be the adjusted opinion of the i th expert and o^g be the group consensus opinion. The soft consensus model of minimum cost based on GWAO under the γ level (the MCC-GWAO- γ model) is defined as:

$$\begin{aligned} \min \phi &= \sum_{i=1}^m c_i |o_i^g - o_i| \\ \text{s.t.} \quad &\begin{cases} 1 - \frac{|o_i^g - o_i|}{o_i^g} \geq \gamma & (7-1) \\ o^g = \prod_{i=1}^m o_i^{w_i} & (7-2) \\ o_i^g, o^g \geq 0 \end{cases} \end{aligned} \quad (7)$$

- (c) The third model uses harmonic average operator to assemble the consensus opinion of a group. Let $o_i^h, i \in M$ be the adjusted opinion of the i th expert and o^h be the group consensus opinion. The soft consensus model of minimum cost based on HWAO under the γ level (the MCC-HWAO- γ

model) is defined as:

$$\begin{aligned} \min \phi &= \sum_{i=1}^m c_i |o_i^h - o_i| \\ \text{s.t.} \quad &\begin{cases} 1 - \frac{|o_i^h - o_i|}{o_i^h} \geq \gamma & (8-1) \\ o^h = \frac{1}{\sum_{i=1}^m \frac{w_i}{o_i^h}} & (8-2) \\ o_i^h, o^h \geq 0 \end{cases} \end{aligned} \quad (8)$$

According to the above definitions, we can get [Theorem 1](#):

Theorem 1. At any stage of a consensus process in the three basic MCC- γ models, if $o_1^a = o_1^g = o_1^h, o_2^a = o_2^g = o_2^h, \dots, o_m^a = o_m^g = o_m^h$, there must be $o^a \geq o^g \geq o^h$, and the equality holds if and only if $o_1^a = o_1^g = o_1^h = o_2^a = o_2^g = o_2^h = \dots = o_m^a = o_m^g = o_m^h$.

[Theorem 1](#) is a general property in a consensus process measured by the three MCC- γ models. When each expert obtained the same value in the three MCC- γ models, MCC- γ model with the AWAO generates the largest group consensus opinion value. The use of the HWAO obtains the lowest group consensus opinion value. Only when the adjusted opinions of all experts are the same, the group consensus opinion values obtained by the three MCC- γ models are equal.

This section establishes three different soft minimum cost consensus models as examples. The three models are based on three frequently and commonly used WA operators. The difference among the three models is that experts' opinions are aggregated in different ways, which have different meanings and suitable for three common scenarios. The MCC-AWAO- γ model is appropriate for assembling experts' opinions that can be directly added together. It is concise and easy to apply in practice. The MCC-GWAO- γ model aims at dealing with a case, in which the cumulative value of experts' opinions is equal to the sum of the continuous product of opinions that have been aggregated. It embodies the multiplicative relationship among the opinions being assembled. A common application of this in reality is the calculation of the average annual interest rate under compound interest. The MCC-HWAO- γ model embodies an additive relationship among the reciprocals of experts' opinions being aggregated. Since HWAO is actually a reciprocal of the arithmetic mean of the reciprocals of the items being averaged, which actually is a special AWAO. It is designed to deal with the reciprocals of each expert's opinion presented the form of additive relationship, which cannot directly use MCC-AWAO- γ model. In reality, applications such as the mean calculation of velocity, calculation of mixed growth rate, and calculation of mixed prices, should adopt the method of HWAO.

In practical applications, the choice of aggregation operator must be consistent with semantically important properties of the aggregation procedure, and fit the desired data ([Beliakov, Pradera, & Calvo, 2007](#)). In this study, the choice of the three models depends on the meaning of the aggregation operator, which is defined mainly on the semantics of individuals' opinions before aggregating and the obtained collective opinion after aggregating, and the intrinsic relationship between them under a particular situation. In summary, the MCC-AWAO- γ model is used to represent an additive semantics among experts' opinions and the collective opinion is a direct description of the additive average of the group. The MCC-GWAO- γ model should be used when experts' opinions need to reflect multiplied relationship in reality. If the reciprocals of experts' opinions represent an added relationship, the method of AWAO cannot be directly used and we should adopt the MCC-HWAO- γ model.

This paper mainly focuses on the minimum cost consensus model with the most intuitive and widely used arithmetic weighted aggregation operator (The MCC-AWAO- γ model), and the maximum return model is constructed through its dual model. These two kinds of soft consensus cost models are studied and discussed in detail.

3.2. Economic interpretation of MCC- γ model

This section studies the properties and significance of the MCC- γ model with AWAO from a moderator's perspective, and explores the relationships between the optimal experts' opinions and the group's optimal opinion under a certain level of consensus. The interpretation here can be applied to the other two models (the MCC-GWAO- γ model and MCC-HWAO- γ model) to get similar conclusions.

In the MCC-AWAO- γ model, for the objective function $f_i(o'_i) = |o'_i - o_i|$, $i \in M$, there must exist $u_i \geq 0$, $v_i \geq 0$, and $u_i * v_i = 0$, such that $|o'_i - o_i| = u_i + v_i$, $o'_i - o_i = u_i - v_i$. The variables u_i and v_i are introduced and the product $*$ is defined in order to facilitate converting the MCC-AWAO- γ model to a linear form. Only if product $*$ is defined, the MCC-AWAO- γ model can be equivalent to the following formula (9). If $u_i = [|o'_i - o_i| + (o'_i - o_i)]/2$, $v_i = [|o'_i - o_i| - (o'_i - o_i)]/2$, then the above conditions hold. The linear programming format of the nonlinear MCC-AWAO- γ is as follows:

$$\begin{aligned} \text{LPM : } \min \phi &= \sum_{i=1}^m (c_i u_i + c_i v_i) \\ \text{s.t. } &\begin{cases} o'_i - u_i + v_i = o_i, & i \in M & (9-1) \\ o'_i - (2-\gamma)o' \leq 0, & i \in M & (9-2) \\ o'_i - \gamma o' \geq 0, & i \in M & (9-3) \\ \sum_{i=1}^m w_i o'_i - o' = 0 & & (9-4) \\ o'_i, o' \geq 0, & u_i \geq 0, v_i \geq 0, & i \in M \end{cases} \end{aligned} \quad (9)$$

Model (9) constructs the minimum cost consensus under γ level for a group from the perspective of a moderator. The objective function $\phi = \sum_{i=1}^m (c_i u_i + c_i v_i)$ represents the minimum total cost of consensus for a group. The restriction condition (9-1) denotes the limits of the deviation between an expert's original opinion and adjusted opinion. The constraint conditions (9-2) and (9-3) indicate that the group consensus achieves at a certain level of consensus and the limits of the deviation between an expert's adjusted opinion and the group's optimal opinion, respectively. Constraint (9-4) shows that the model uses AWAO to assemble the opinions of experts, and obtain the consensus opinion of a group. Under these constraints, the total cost of achieving group consensus is as small as possible.

Property 3. In the minimum cost model, let $[o_{il}^*, o_{ir}^*]$ denotes the range of the deviation for a given consensus level, then $o^* = (o_{il}^* + o_{ir}^*)/2 = \sum_{i=1}^m w_i o_i^* / m$ holds.

Theorem 2 further reveals the relations between experts' optimal opinions and the group optimal opinion under a certain consensus degree.

Theorem 2. In the minimum cost model, let o_i^* be the optimal opinions of experts ($e_i \in E$), and o^* be the group optimal opinion under a

certain consensus degree. Then,

$$\begin{cases} \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) \\ = \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] & m = 2n, i = 1, 2, \dots, n \quad (1) \\ \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) + \frac{(2-\gamma)o^* - \gamma o^*}{2m} \\ = \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] & m = 2n+1, i = 1, 2, \dots, n \quad (2) \end{cases}$$

hold.

Proof. We have

$$\begin{aligned} &\sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) - \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] \\ &= \sum_{i=1}^n w_i o_i^* + \sum_{i=n+1}^m w_i o_i^* - \sum_{i=1}^n \frac{\gamma o^*}{m} - \sum_{i=n+1}^m \frac{2o^*}{m} + \sum_{i=n+1}^m \frac{\gamma o^*}{m} \\ &= \sum_{i=1}^m w_i o_i^* - \frac{\gamma}{m} \left(\sum_{i=1}^n o^* - \sum_{i=n+1}^m o^* \right) - \frac{2}{m} \sum_{i=n+1}^m o^* \\ (1) \text{ When } m = 2n, i = 1, 2, \dots, n, \text{ we have } \sum_{i=1}^n o^* &= \sum_{i=n+1}^m o^* \\ \text{and } \sum_{i=n+1}^m o^* = \frac{m}{2} o^* \text{ hold. Thus } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) - \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] &= \sum_{i=1}^n w_i o_i^* - o^* \text{ hold, according to Property 3, we have } \sum_{i=1}^m w_i o_i^* = o^* \text{ hold. Therefore, we have } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) - \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] = 0 \text{ hold. Thus, } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) = \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] \quad m = 2n, i = 1, 2, \dots, n \text{ hold.} \\ (2) \text{ When } m = 2n+1, i = 1, 2, \dots, n, \text{ we have } \sum_{i=n+1}^m o^* - \sum_{i=1}^n o^* = o^* \text{ and } \sum_{i=n+1}^m o^* = \frac{m+1}{2} o^* \text{ hold. Thus, } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) - \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] &= \sum_{i=1}^n w_i o_i^* + \frac{\gamma}{m} o^* - \frac{m+1}{2} o^* \text{ hold, according to Property 3, we have } \sum_{i=1}^m w_i o_i^* = o^* \text{ hold. Therefore, we have } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) - \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] = o^* + \frac{\gamma}{m} o^* - \frac{m+1}{2} o^* = -\frac{1-\gamma}{2m} o^*. \text{ Since } \frac{1-\gamma}{m} o^* = \frac{(2-\gamma)o^* - \gamma o^*}{2m}, \text{ thus } \sum_{i=1}^n (w_i o_i^* - \frac{\gamma o^*}{m}) + \frac{(2-\gamma)o^* - \gamma o^*}{2m} = \sum_{i=n+1}^m [\frac{(2-\gamma)o^*}{m} - w_i o_i^*] \quad m = 2n+1, i = 1, 2, \dots, n \text{ hold. } \square \end{aligned}$$

To reveal the meaning of Theorem 2, we consider the case of equal weights of individual opinions. We can get Corollary 1.

Corollary 1. In the minimum cost model, if $w_1 = w_2 = \dots = w_m = 1/m$, Then,

$$\begin{cases} \sum_{i=1}^n (o_i^* - \gamma o^*) = \sum_{i=n+1}^m [(2-\gamma)o^* - o_i^*] & m = 2n, i = 1, 2, \dots, n \\ \sum_{i=1}^n (o_i^* - \gamma o^*) + \frac{(2-\gamma)o^* - \gamma o^*}{2} \\ = \sum_{i=n+1}^m [(2-\gamma)o^* - o_i^*] & m = 2n+1, i = 1, 2, \dots, n \end{cases}$$

hold.

Corollary 1 explains the relationships between an expert's optimal opinion and the group optimal opinion under a certain degree of consensus. When the number of experts is even, the sum of the deviations between the minimum value of the optimal group opinion and one part of experts' optimal opinion must be equal to the sum of deviations between the maximum value of the optimal group opinion and the rest part of experts' optimal opinion. When the number of experts is odd under a certain degree of consensus, the above-mentioned two parts of the deviations are not

equal, and the difference between the two parts is equal to half the difference between the maximum and minimum value of the group optimal opinion.

The following example further interprets the minimum cost consensus model under a certain level in GDM. The data were from Ref. Ben-Arieh et al. (2009).

Suppose there are four experts e_1, e_2, e_3, e_4 in a GDM and their corresponding opinions are $o_1 = 0, o_2 = 3, o_3 = 6, o_4 = 10$, and the weights of their opinions are $w_1 = 0.3, w_2 = 0.1, w_3 = 0.4, w_4 = 0.2$, respectively. Suppose that the unit costs paid to the four experts by a moderator are $c_1 = 1, c_2 = 4, c_3 = 3, c_4 = 5$, respectively. After consultation and discussion, the consensus level of the group consensus cannot be lower than 0.8, namely, $\gamma = 0.8$.

Let the consensus opinion of the group be o' and an expert's final opinion after several adjustments is $o'_i, i = 1, 2, 3, 4$. Using AWAO to assemble the consensus opinion of the group, we construct the optimal consensus decision making model based on minimum cost under a certain degree of consensus as follows:

$$PM_1: \min \phi = |o'_1 - 0| + 4|o'_2 - 3| + 3|o'_3 - 6| + 5|o'_4 - 10|$$

$$s.t. \begin{cases} 1 - \frac{|o'_1 - o'|}{o'_1} \geq 0.8 \\ 1 - \frac{|o'_2 - o'|}{o'_2} \geq 0.8 \\ 1 - \frac{|o'_3 - o'|}{o'_3} \geq 0.8 \\ 1 - \frac{|o'_4 - o'|}{o'_4} \geq 0.8 \\ o' = 0.3o'_1 + 0.1o'_2 + 0.4o'_3 + 0.2o'_4 \\ o'_1, o'_2, o'_3, o'_4, o' \geq 0 \end{cases} \quad (10-1)$$

(10)

By letting $|o'_i - o_i| = u_i + v_i, o'_i - o_i = u_i - v_i, i = 1, 2, 3, 4$. Model (10) is equivalent to the following linear programming model:

$$LPM_1: \min \phi = u_1 + v_1 + 4u_2 + 4v_2 + 3u_3 + 3v_3 + 5u_4 + 5v_4$$

$$s.t. \begin{cases} o'_1 - u_1 + v_1 = 0 \\ o'_2 - u_2 + v_2 = 3 \\ o'_3 - u_3 + v_3 = 6 \\ o'_4 - u_4 + v_4 = 10 \\ o'_1 - 1.2o' \leq 0 \\ o'_2 - 1.2o' \leq 0 \\ o'_3 - 1.2o' \leq 0 \\ o'_4 - 1.2o' \leq 0 \\ o'_1 - 0.8o' \geq 0 \\ o'_2 - 0.8o' \geq 0 \\ o'_3 - 0.8o' \geq 0 \\ o'_4 - 0.8o' \geq 0 \\ 0.3o'_1 + 0.1o'_2 + 0.4o'_3 + 0.2o'_4 - o' = 0 \\ o'_i, o', u_i, v_i \geq 0, i = 1, 2, 3, 4 \end{cases} \quad (11-1)$$

(11-2)

(11-3)

(11-4)

(11-5)

(11)

The unique solution to Model (11) is $\bar{X}^* = (9.0 \ 3.0 \ 0.0 \ 0.0 \ 1.9 \ 6.6 \ 9.7 \ 5)^T$, and the optimal value of the objective function of Model (11) is $\min \phi = 26$. The optimal consensus opinions of the four experts are $o_1^* = 9, o_2^* = 6, o_3^* = 6, o_4^* = 9$, and the optimal consensus opinion for the group is $o^* = 7.5$, which satisfies the expected level $\gamma = 0.8$.

3.3. Example

This section further analyzes the changes in experts' opinions and the consumption of resources at different consensus levels using the three consensus models (The MCC-AWAO- γ , MCC-GWAO- γ and MCC-HWAO- γ models). The results of the three models are summarized in Tables 1–3.

It is obvious that the higher the consensus level in GDM, the greater the cost of the group consensus, which is increased in an exponential form. Figs. 1–3 reflect the changes of experts' optimal

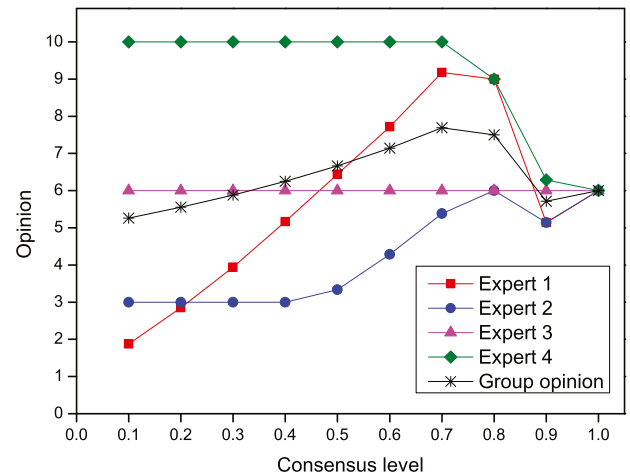


Fig. 1. The changes of experts' optimal opinions and group's optimal opinions of MCC-AWAO- γ model.

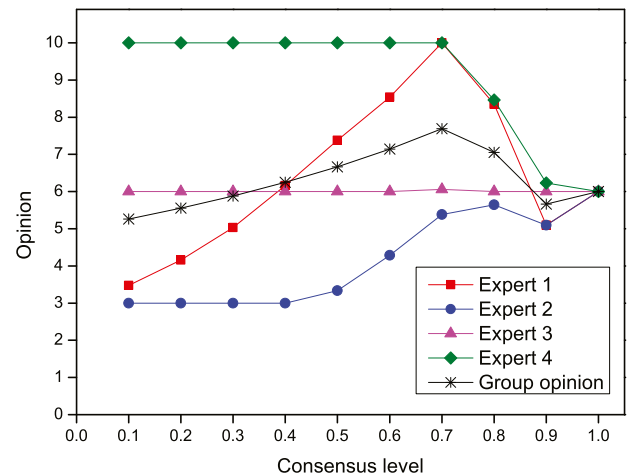


Fig. 2. The changes of experts' optimal opinions and group's optimal opinions of MCC-GWAO- γ model.

opinions and the optimal group opinion at different consensus levels (degree of convergence of opinion). Fig. 4 shows the changes of the total consensus costs in the three models under different consensus levels.

We observe that different aggregation operators can affect the opinions of experts, but the convergence degrees of the experts' opinions as a whole is quite similar. Moreover, the costs of consensus models constructed with different weighted operators are relatively close under the same consensus level, which means the impact of different weighted operators on the consensus cost is relatively small. The convergence of experts' opinions and the consensus costs are highly correlated with the consensus level. At different consensus levels, the magnitude of experts' change of optimal opinions are different. In general, the higher the group consensus level is, the more opinion change is required, which indicates that a moderator needs to persuade more experts to change their opinions. When complete agreement is reached ($\gamma = 1$), the optimal opinions of experts are equal to the optimal group consensus opinion. The lower the consensus level of a group, the less opinion change of experts is needed. Most experts can keep their original opinions.

Table 1Result of the MCC-AWAO- γ model under different consensus levels.

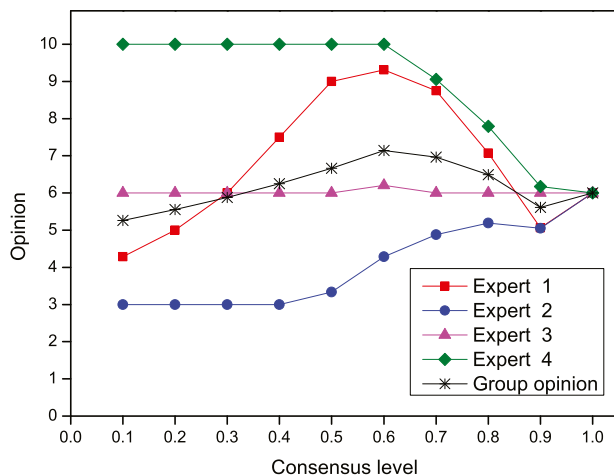
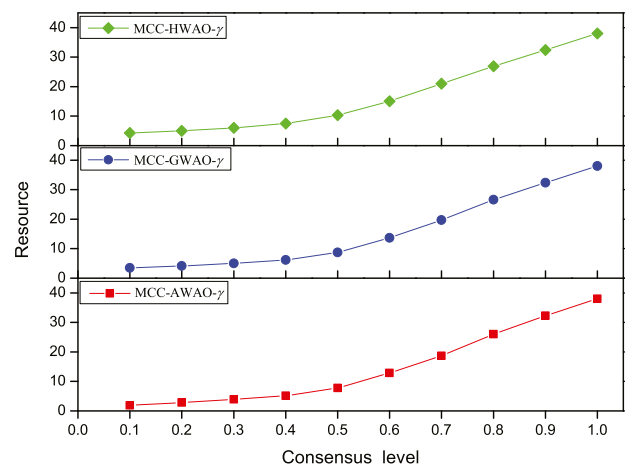
	$\gamma = 0.1$	$\gamma = 0.2$	$\gamma = 0.3$	$\gamma = 0.4$	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$	$\gamma = 0.9$	$\gamma = 1.0$
σ_1^*	1.8772	2.8519	3.9412	5.1667	6.4444	7.7143	9.1795	9	5.1429	6
σ_2^*	3	3	3	3	3.3333	4.2857	5.3846	6	5.1429	6
σ_3^*	6	6	6	6	6	6	6	6	6	6
σ_4^*	10	10	10	10	10	10	10	9	6.2857	6
σ^*	5.2632	5.5556	5.8824	6.25	6.6667	7.1429	7.6923	7.5	5.7143	6
ϕ	1.877	2.852	3.941	5.167	7.777	12.857	18.717	26	32.286	38

Table 2Result of the MCC-GWAO- γ model under different consensus levels.

	$\gamma = 0.1$	$\gamma = 0.2$	$\gamma = 0.3$	$\gamma = 0.4$	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$	$\gamma = 0.9$	$\gamma = 1.0$
σ_1^*	3.4747	4.1609	5.0342	6.1616	7.3768	8.5382	10	8.3518	5.0972	6
σ_2^*	3	3	3	3	3.3333	4.2857	5.3846	5.6438	5.0972	6
σ_3^*	6	6	6	6	6	6	6.0583	6	6	6
σ_4^*	10	10	10	10	10	10	10	8.4658	6.2299	6
σ^*	5.2632	5.5556	5.8824	6.25	6.6667	7.1429	7.6923	7.0548	5.6636	6
ϕ	3.4747	4.1609	5.0342	6.1616	8.7101	13.681	19.7135	26.5983	32.3364	38

Table 3Result of the MCC-HWAO- γ model under different consensus levels.

	$\gamma = 0.1$	$\gamma = 0.2$	$\gamma = 0.3$	$\gamma = 0.4$	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$	$\gamma = 0.9$	$\gamma = 1.0$
σ_1^*	4.2857	5	6	7.5	9	9.3103	8.7532	7.0704	5.0624	6
σ_2^*	3	3	3	3	3.3333	4.2857	4.8772	5.1942	5.0534	6
σ_3^*	6	6	6	6	6	6.2069	6	6	6	6
σ_4^*	10	10	10	10	10	10	9.0577	7.7912	6.1764	6
σ^*	5.2632	5.5556	5.8824	6.25	6.6667	7.1429	6.9675	6.4927	5.6149	6
ϕ	4.2857	5	6	7.5	10.3333	15.0739	20.9735	26.8909	32.394	38

**Fig. 3.** The changes of experts' optimal opinions and group's optimal opinions of MCC-HWAO- γ model.**Fig. 4.** Changes of the total consensus costs in the three Models under different consensus levels.

4. Maximum return model under a certain consensus level

The MCC- γ models in Section 3 construct the cost of consensus under a certain level of consensus from a moderator's point of view. We study the properties and significance of the models, relationships between the optimal experts' opinions and the group's optimal opinion, and discuss the convergence of individuals' opinions and the cost of resources in different WA operators under a certain level of consensus.

However, consensus is a complex and resource-consuming process, which requires the exchange of opinions between a moderator and experts (Gong, Xu, Li, et al., 2015; Gong, Zhang et al., 2015; Asongu and Nwachukwu, 2017). Therefore, the study of compensa-

tion that an expert expected for changing his/her opinion is necessary. We need to clarify the issues under a certain level of consensus from experts' perspective: (1) The composition of an expert's expected unit compensation. (2) The relationships between the total compensation expected by experts and the total cost paid by a moderator when the consensus level has been reached (3) Relationships between the expected unit compensation of an expert and the adjustment of his/her opinion. (4) Relationships between the compositions of an expert's expected unit compensation and his/her adjusted opinion. Through the dual model of the minimum cost consensus model, we construct the maximum return model from experts' angle under a certain level of consensus.

The maximum return model under a certain level of consensus can be constructed through the duality of model (9). According

to the duality theory of linear programming, the duality of model (9) can be expressed as follows:

$$DLPM(I): \max \psi = \sum_{i=1}^m o_i y_i$$

$$\text{s.t.} \begin{cases} -y_i \leq c_i, & i \in M & (12-1) \\ y_i \leq c_i, & i \in M & (12-2) \\ y_i + y_{i+m} + y_{i+2m} + w_i x \leq 0, & i \in M & (12-3) \\ \sum_{i=1}^m [-(2-\gamma)y_{i+m} - \gamma y_{i+2m}] - x \leq 0 & (12-4) \\ y_{i+m} \leq 0, y_{i+2m} \geq 0 \end{cases}$$

Obviously, $o_i^* > 0$, $o^* > 0$. Based on the complementary slackness property of dual linear programming theory, there exists $y_i + y_{i+m} + y_{i+2m} + w_i x = 0$, $i \in M$ and $\sum_{i=1}^m [-(2-\gamma)y_{i+m} - \gamma y_{i+2m}] - x = 0$ holds. Thus, Model (12) is equivalent to

$$DLPM(II): \max \psi = \sum_{i=1}^m o_i y_i$$

$$\text{s.t.} \begin{cases} |y_i| \leq c_i, & i \in M & (13-1) \\ y_i + y_{i+m} + y_{i+2m} + w_i x = 0, & i \in M & (13-2) \\ \sum_{i=1}^m [-(2-\gamma)y_{i+m} - \gamma y_{i+2m}] - x = 0 & (13-3) \\ y_{i+m} \leq 0, y_{i+2m} \geq 0 \end{cases}$$

Model (13) is also the dual problem of Model (9). It is easy to prove that the set of feasible solutions $\bar{Y} = (y_1, \dots, y_{3m}, x)^T$ of Model (12) or model (13) is nonempty. The number of basic feasible solutions is finite.

In order to achieve group consensus, experts need to make sacrifice and change their opinions. Accordingly, they expect compensations and always want to maximize their returns. Model (13) reveals the economic meaning from the angle of individual expert, and constructs the compensation model based on personal interest maximization.

The following subsections analyze sufficient conditions of the maximum return model, explain the economic significance of the maximum return model, discuss its relationship with the minimum consensus cost models, and use a numerical example to show the relationship between these two kinds of models.

4.1. Two sufficient conditions of maximum return model

In order to explain the economic significance of variable y_i in the maximum return model, we construct the sufficient conditions that (a) the unit compensation of an expert ($|y_i|$) is equal to the unit payment c_i of a moderator, (b) the adjusted optimal opinion of an expert o_i^* is equal to the original opinion o_i , and explore important properties of the maximum return model of experts under a certain degree of consensus.

Theorem 3. If $o_i^* > o_i$, then $y_i = -c_i < 0$; if $o_i^* < o_i$, then $y_i = c_i > 0$; if $o_i^* = o_i$, then $-c_i \leq y_i \leq c_i$.

Proof. If $o_i^* > o_i$, then $o_i^* - o_i = u_i - v_i > 0$, and $u_i > v_i$. Because $u_i \geq 0$, $v_i \geq 0$, $u_i * v_i = 0$, we have $v_i = 0$, $u_i > 0$. According to the principle of complementary slackness, we have $y_i = -c_i$; likewise, if $o_i^* < o_i$, we have $y_i = c_i > 0$; if $o_i^* = o_i$, we also have $-c_i \leq y_i \leq c_i$. $\square$

Theorem 3 implies when the desired consensus degree is reached and an expert's adjusted optimal opinion is higher than his/her original opinion, the expert's expected unit compensation

(shadow profit) is negative. On the other hand, if an expert's adjusted optimal opinion is lower than his/her initial opinion, the expert's expected unit compensation (shadow profit) is positive. The absolute value of an expert's expected unit compensation equals to the actual unit compensation cost paid by a moderator.

Two important corollaries of the theory in a certain degree of consensus can be drawn from **Theorem 3**:

Corollary 2. When the desired consensus degree is reached, for any expert e_{i_0} , if the opinion satisfies $o_{i_0} \neq o_{i_0}^*$, then the equation $|y_{i_0}| = c_{i_0}$ holds.

Corollary 2 indicates that when an expert arrives at the desired level in a certain degree of consensus, if he/she adjusts his/her initial opinion, he/she will get a certain amount of compensation and the compensation is equal to the cost paid by the moderator. This gives the sufficient condition that an expert obtained the fully payment by the moderator in a certain degree of consensus from the mathematical aspect.

Corollary 3. For all experts, if the opinions satisfy $o_i^* \neq o_i$, $i \in M$, then $|y_i| = c_i$, $i \in M$ holds.

Corollary 3 is a special case that in order to reach group consensus with expected level, all experts adjusted and changed their initial opinions. The moderator needs to pay the maximum compensation to satisfy $|y_i| = c_i$, $i \in M$ to achieve a consensus. When the expected level of consensus is high, this situation can happen.

Theorem 4. If $y_i = c_i$, then $o_i^* \leq o_i$; if $y_i = -c_i$, then $o_i^* \geq o_i$; and if $-c_i < y_i < c_i$, then $o_i^* = o_i$.

Proof. If $y_i = c_i$, then there must be $-y_i < c_i$. According to the principle of complementary slackness, we have $u_i = 0$, and thus $o_i^* = o_i - v_i \leq o_i$. If $y_i = -c_i$, then there must be $y_i < c_i$. According to the principle of complementary slackness, we have $v_i = 0$, and thus $o_i^* = o_i + u_i \geq o_i$. If $-c_i < y_i < c_i$, according to the principle of complementary slackness, we have $u_i = v_i = 0$, and thus $o_i^* = o_i$ holds. $\square$

Theorem 4 implies that if an expert's expected unit compensation is positive and equals to the unit cost paid by a moderator, then the expert's adjusted optimal opinion must not exceed his/her original opinion. If an expert's expected unit compensation is negative and the absolute value of shadow profit is equal to the unit cost paid by a moderator, the expert's adjusted optimal opinion must not be lower than his/her original opinion. If the absolute value of an expert's expected unit compensation is lower than the unit cost paid by a moderator, then his/her adjusted optimal opinion is equal to the original opinion.

Theorem 4 can be used to determine whether an expert has adjusted his/her original opinion by calculating the shadow price of each expert in the minimum cost model. If the unit compensation expected by an expert is below the unit compensation paid by a moderator, the expert does not adjust his/her opinion. It is a sufficient condition that an expert's adjusted opinion is equal to his/her initial opinion, and draws the following conclusion:

Corollary 4. When the desired consensus degree is reached, for any expert i_0 , if the strict inequality $|y_{i_0}| < c_{i_0}$ holds, then $o_{i_0}^* = o_{i_0}$.

Corollary 4 indicates that when the unit compensation expected by an expert is lower than the unit compensation paid by a moderator, the moderator does not need to persuade him/her to change his/her opinion. The moderator can give less unit compensation to an expert than the initial unit payment to keep the expert's initial opinion unchanged.

Corollary 5. For all experts, if $|y_i| < c_i$, $i \in M$, then $o_i^* = o_i$, $i \in M$ holds.

Corollary 5 is a special case in which the unit compensations expected by all experts are lower than the cost paid by a moderator, indicating that all experts do not adjust their opinions when the consensus is reached. This can happen in two situations: the expected consensus level γ is relatively low; and the rate of convergence of experts' initial opinions is relatively high.

4.2. Economic interpretation of the maximum return model

This section analyzes the economic interpretations of the variables, constraints, and objective function of the maximum return model, and further reveals the composition of an expert's expected unit compensation under a certain consensus level. Moreover, when a consensus level is reached, the relationship between the total compensation expected by experts and the total compensation paid by a moderator is also explored.

4.2.1. Economic interpretations of the variables

In Model (13), the variable y_i is the unit return (compensation) that expert e_i expects to obtain for changing his/her original opinion. From Theorems 3 and 4, we can see that negative y_i^- means that an expert's adjusted opinion is not lower than the original opinion when the consensus degree is reached. That is, expert e_i should adjust his/her opinion in the increased direction to achieve a consensus. A positive y_i^+ means that an expert's adjusted opinion is not higher than the original opinion when a consensus degree is reached, that is, the expert e_i should adjust the opinion in the decreased direction to achieve the consensus. Regardless of the signs of y_i , we always use $|y_i|$ to represent the expected unit compensation for individual expert.

An expert's expected unit compensation $|y_i|$ includes two parts:

- The first part of compensation: satisfying the level of group consensus. In order to obtain the desired group consensus γ , the difference between an expert's adjusted opinion o_i^* and the group's optimal opinion o^* should be within a certain range, which is accepted by the whole group. The range of the deviation can be denoted by $[o_{il}^*, o_{ir}^*]$. Only when $o_i^* \in [o_{il}^*, o_{ir}^*]$, the desired consensus can be achieved. Experts have to adjust their initial opinions in the deviation range $[o_{il}^*, o_{ir}^*]$, and therefore expect compensation.
- The second part of compensation: contributing to the formation of an optimal group consensus opinion. An expert contributes to the final group consensus by changing his/her opinion to get close to the optimal consensus opinion, and thus needs to obtain certain compensation. In other words, an expert's optimal opinion satisfies the conditions of a given consensus level, also forms the optimal group opinion, which is the best for the whole group.

In Model (13), the variable y_{i+2m} , $i \in m$ represents the unit compensation for an expert when his/her adjusted opinion meets $o_i^* > o_{il}^*$. The variable y_{i+m} , $i \in M$ represents the unit compensation for an expert when his/her adjusted opinion satisfies $o_i^* < o_{ir}^*$. The sum of y_{i+2m} , $i \in M$ and y_{i+m} , $i \in M$ is the unit compensation that an expert should be paid because his/her adjusted opinion is in the accepted consensus range, i.e., the first part of unit compensation. The variable x means that all experts contribute to form the ideal (optimal) group opinion to reach the consensus and experts obtained the total unit compensation. $w_i x$ indicates that an expert e_i contributed to the ideal group opinion by adjusting his/her opinion and obtained the unit compensation, i.e., the second part of unit compensation. The economic interpretations of the variables in the maximum return model are shown in Fig. 5.

4.2.2. Economic interpretation of the restrictions

There are three constraints in model (13). Constraint (13-1) indicates the relationship between the unit cost paid by a moderator and the unit compensation expected by experts. Constraint (13-2) reveals the compositions of the expected unit compensation of experts. Constraint (13-3) indicates the relationship between a group consensus level and group compensations. The economic significance is as follows.

In Model (13), the restriction (13-1) indicates that an expert's expected compensation $|y_i|$ cannot exceed the unit compensation that a moderator is willing to pay. It intuitively reflects the relationship between the unit compensation paid by a moderator and the unit compensation that experts expected in real decision scenarios.

The constraint (13-2) is equivalent to $|y_i| = |y_{i+m} + y_{i+2m} + w_i x|$, which reveals that an expert's expected compensation y_i has two parts. The first part ($y_{i+m} + y_{i+2m}$) is the unit compensation that each unit opinion change by an expert contributes to a desired level of consensus. The second part ($w_i x$) is the unit compensation that each unit opinion change by an expert contributes to the formation of an ideal (optimal) group opinion.

The constraint (13-3) indicates that there is a special numerical relationship between a group consensus level and group compensations. To clarify such relationship, we will prove that the constraint (13-3) is equivalent to the following Eq. (14).

$$\gamma = \frac{\sum_{i=1}^m (y_{i+m} - y_{i+2m}) - \sum_{i=1}^m y_i}{\sum_{i=1}^m (y_{i+m} - y_{i+2m})} \quad (14)$$

According to constraint (13-2), we have $\sum_{i=1}^m (y_i + y_{i+m} + y_{i+2m}) + \sum_{i=1}^m w_i x = 0$. Because $\sum_{i=1}^m w_i = 1$, the equation $\sum_{i=1}^m (y_i + y_{i+m} + y_{i+2m}) = -x$ holds. Constraint (13-3) is equivalent to $\sum_{i=1}^m [-(2 - \gamma)y_{i+m} - \gamma y_{i+2m}] + \sum_{i=1}^m (y_i + y_{i+m} + y_{i+2m}) = 0$, and we have $\gamma \sum_{i=1}^m y_{i+m} - \gamma \sum_{i=1}^m y_{i+2m} + \sum_{i=1}^m y_i - \sum_{i=1}^m y_{i+m} + \sum_{i=1}^m y_{i+2m} = 0$. Thus, we can get the Eq. (14).

Eq. (14) expresses the relationship between a group consensus degree and the unit compensations in the maximum return consensus model, and indicates that the unit compensations of experts expressed using proportion can reflect the degree of consensus in the maximum return model.

From the above discussion, two properties of the maximum return consensus model under a certain degree of consensus can be obtained and described as follows.

Property 4. In the maximum return consensus model under a certain degree of consensus, the variables of compensations y_i, y_{i+m}, y_{i+2m} and x satisfy $|y_i| = |y_{i+m} + y_{i+2m}| + |w_i x|$, $i \in M = \{1, 2, \dots, m\}$.

Property 5. In the maximum return consensus model under a certain degree of consensus, let y_i, y_{i+m} and y_{i+2m} be defined as above, the group consensus degree is denoted as γ , there must have $\gamma = \frac{\sum_{i=1}^m (y_{i+m} - y_{i+2m}) - \sum_{i=1}^m y_i}{\sum_{i=1}^m (y_{i+m} - y_{i+2m})}$.

4.2.3. Economic interpretation of the objective function

To interpret the meaning of the objective function of Model (13), we obtain the following Theorem 5.

Theorem 5. when ψ obtains maximum value, the objective function $\sum_{i=1}^m o_i y_i$ of Model (13) is equal to $\sum_{i=1}^m y_i (o_i - o_i^*)$.

Proof. Since $\sum_{i=1}^m y_i (o_i - o_i^*) - \sum_{i=1}^m o_i y_i = -\sum_{i=1}^m y_i o_i^*$, we need to prove the equality $-\sum_{i=1}^m y_i o_i^* = 0$ holds. According to the principle of complementary slackness and (13), $o_i^* (y_i + y_{i+m} + y_{i+2m} + w_i x) = 0$ holds, then,

$$-\sum_{i=1}^m y_i o_i^* = \sum_{i=1}^m o_i^* y_{i+m} + \sum_{i=1}^m o_i^* y_{i+2m} + x \sum_{i=1}^m w_i o_i^* \quad (15)$$

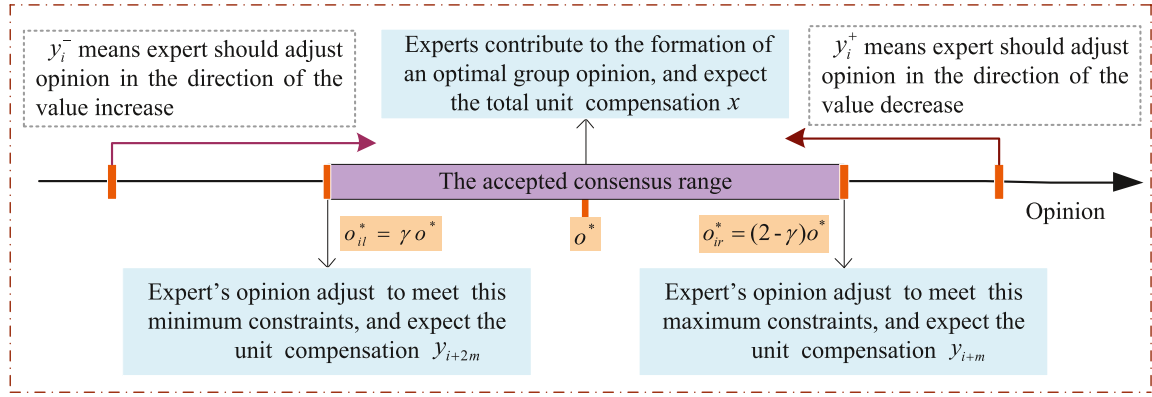


Fig. 5. Economic interpretations of the variables in the maximum return model.

According to the principle of complementary slackness and (9-2), (9-3), $o_i^* y_{i+m} = (2 - \gamma) o^* y_{i+m}$, $o_i^* y_{i+2m} = \gamma o^* y_{i+2m}$ holds. Then, in (15)

$$-\sum_{i=1}^m y_i o_i^* = \sum_{i=1}^m (2 - \gamma) o^* y_{i+m} + \sum_{i=1}^m \gamma o^* y_{i+2m} + x \sum_{i=1}^m w_i o_i^* \quad (16)$$

According to the principle of complementary slackness and (13), $o^* \{ \sum_{i=1}^m [-(2 - \gamma) y_{i+m} - \gamma y_{i+2m}] - x \} = 0$. Then, $o^* \sum_{i=1}^m (2 - \gamma) y_{i+m} + o^* \sum_{i=1}^m \gamma y_{i+2m} + o^* x = 0$ holds, and from (9), $x \sum_{i=1}^m w_i o_i^* - x o^* = 0$ holds. The above Eq. (16) is equal to

$$-\sum_{i=1}^m y_i o_i^* = \sum_{i=1}^m (2 - \gamma) o^* y_{i+m} + \sum_{i=1}^m \gamma o^* y_{i+2m} + x o^* = 0$$

So the original equation $\sum_{i=1}^m o_i y_i = \sum_{i=1}^m y_i (o_i - o_i^*)$ holds. $\square$

Theorem 5 explains the significance of the objective function in the maximum return model. The maximum expected compensation obtained by all experts is $\sum_{i=1}^m y_i (o_i - o_i^*)$, which is equal to the sum of products of the opinion deviations from their original opinions to the adjusted opinions and the corresponding expected compensations.

4.3. Relationships between the soft minimum cost and maximum return models

From the original problem and the dual problem, we can get **Theorems 6** and **7** to explain the relationships between the soft minimum cost and maximum return models. Actually, **Theorems 6** and **7** reveal the relationships between the first part of compensation expected by an expert and his/her adjusted optimal opinion.

Lemma 1. In the maximum return model (13) under a certain consensus level, $y_{i+m} * y_{i+2m} = 0$ holds for $i = 1, 2, \dots, m$.

Proof. If $y_{i+m} = 0$, it is clear that the equation holds.

If $y_{i+m} \neq 0$, according to the principle of complementary slackness, $o_i^* - (2 - \gamma) o^* = 0$ holds. If $o_i^* - \gamma o^* = 0$, then $\gamma o^* = (2 - \gamma) o^*$ and $\gamma = 1$, which is not a soft consensus level, thus we have $o_i^* - \gamma o^* > 0$. According to the principle of complementary slackness, $y_{i+2m} = 0$ holds, and $y_{i+m} * y_{i+2m} = 0$ holds. $\square$

From **Lemma 1**, we know that the value of y_{i+m} and y_{i+2m} can have the following three situations:

- (1) $y_{i+m} < 0$, $y_{i+2m} = 0$.
- (2) $y_{i+m} = 0$, $y_{i+2m} > 0$.
- (3) $y_{i+m} = 0$, $y_{i+2m} = 0$.

According to **Lemma 1** and the principle of complementary slackness, we can obtain the following **Theorem 6** and **Theorem 7**.

Theorem 6. If $o_i^* = \gamma o^*$, then $y_{i+m} = 0$; if $\gamma o^* < o_i^* < (2 - \gamma) o^*$, then $y_{i+m} = 0$, $y_{i+2m} = 0$; and if $o_i^* = (2 - \gamma) o^*$, then $y_{i+2m} = 0$.

Theorem 6 reveals the relationships between the value of an expert's adjusted optimal opinion o_i^* ($o_i^* \in [\gamma o^*, (2 - \gamma) o^*]$) and the compensations y_{i+m} and y_{i+2m} for satisfying a given consensus degree.

If an expert's adjusted opinion equals to the minimum of an acceptable consensus range when a group consensus has been reached, the part of expert's expected unit compensation due to the limitation of the maximum value of an acceptable consensus range is 0. If an expert's adjusted opinion is between the minimum and maximum values of an acceptable consensus range, his/her adjusted opinion is not affected by an acceptable range. Thus the part of expert's expected unit compensation due to the limitation of the minimum and maximum values of an acceptable consensus range are both 0. If an expert's adjusted opinion equals to the maximum of an acceptable consensus range when a group consensus has been reached, the part of expert's expected unit compensation value due to the limitation of the minimum value of an acceptable consensus range is 0.

Theorem 7. If $y_{i+m} < 0$, $y_{i+2m} = 0$, then $o_i^* = (2 - \gamma) o^*$ holds; if $y_{i+m} = 0$, $y_{i+2m} > 0$, then $o_i^* = \gamma o^*$ holds; and if $y_{i+m} = 0$, $y_{i+2m} = 0$, then $\gamma o^* \leq o_i^* \leq (2 - \gamma) o^*$ holds.

Theorem 7 describes the relationships of y_{i+m} and y_{i+2m} in the maximum return model with the corresponding o_i^* and o^* in the minimum cost consensus model.

When an expert's adjusted opinion is located in the optimal consensus interval and his/her expected partial (not include the second part of compensation) unit compensation is not 0 due to the limitation of the maximum value of the consensus interval, his/her adjusted opinion must be equal to the maximum value of the acceptable range of opinions when the group consensus is reached. When an expert's adjusted opinion is located in the optimal consensus interval, and his/her expected partial unit compensation is not 0 due to the limitation of the minimum value of the consensus interval, his/her adjusted opinion must be equal to the minimum value of the acceptable range of opinions deviation when the group consensus is reached. When an expert's expected first part of unit compensation is 0, the expert's adjusted opinion is located in the value of the acceptable range of opinions deviation when the group consensus is reached.

4.4. Numerical example

This section uses the same example (Ben-Arieh et al., 2009) in Section 3.2 to show the relationships between the minimum cost consensus model and the maximum return model and validates the theoretical analysis of the above theorems.

The dual problem of model (11) is:

$$DLPM_1: \max \psi = 0y_1 + 3y_2 + 6y_3 + 10y_4$$

$$s.t. \begin{cases} |y_1| \leq 1 \\ |y_2| \leq 4 \\ |y_3| \leq 3 \\ |y_4| \leq 5 \end{cases} \quad (17-1)$$

$$\begin{cases} y_1 + y_5 + y_9 + 0.3x \leq 0 \\ y_2 + y_6 + y_{10} + 0.1x \leq 0 \\ y_3 + y_7 + y_{11} + 0.4x \leq 0 \\ y_4 + y_8 + y_{12} + 0.2x \leq 0 \end{cases} \quad (17-2)$$

$$-x - 1.2y_5 - 1.2y_6 - 1.2y_7 - 1.2y_8 \quad (17-3)$$

$$-0.8y_9 - 0.8y_{10} - 0.8y_{11} - 0.8y_{12} \leq 0 \quad (17-4)$$

$$y_5, y_6, y_7, y_8 \leq 0, y_9, y_{10}, y_{11}, y_{12} \geq 0 \quad (17-5)$$

Model (17) has the unique solution $\bar{Y}^* = (-1 \ -4 \ -2 \ 5 \ -0.5 \ 0 \ 0 \ -6 \ 0 \ 3.5 \ 0 \ 0 \ 5)^T$, and its maximum objective function is $\max \psi = 26$. Both the primal model (LPM_1) and its dual model ($DLPM_1$) have optimal solutions and their optimal objective functions are both 26. Economically, this indicates that the total return of all experts is equal to the total cost paid by a moderator and the optimal consensus has been reached.

Expected unit returns of experts solved by Model (17) are $y_1 = -1$, $y_2 = -4$, $y_3 = -2$, $y_4 = 5$. The positive or negative value represents the opinion direction that an expert should adjust to achieve the consensus. In order to achieve the desired group consensus $\gamma = 0.8$, $y_5 = -0.5$, $y_6 = 0$, $y_7 = 0$ and $y_8 = -6$ are the part of unit compensations expected by experts, because their adjusted opinions are restricted by the maximum value ($o_{ir} = 9$) of the group consensus range. $y_9 = 0$, $y_{10} = 3.5$, $y_{11} = 0$ and $y_{12} = 0$ express the part of the unit compensation expected by the experts, because their adjusted opinions are restricted by the minimum value ($o_{il} = 6$) of the group consensus range. $x = 5$ indicates that all the experts' adjusted opinions contribute to the formation of the group consensus, and should receive unit expected compensation. The total unit compensation should be 5.

In model (17), because the adjusted optimal opinions of expert 1 (expert 2) is higher than his/her original opinion, the expected unit compensation $y_1 = -1$ ($y_2 = -4$) is equal to the opposite value of the moderator's unit cost $c_1 = 1$ ($c_2 = 4$). Likewise, expert 4's adjusted optimal opinion is lower than his/her original opinion, then the expected unit compensation $y_4 = 5$ is equal to the value of the moderator's unit cost $c_4 = 5$. (The economic significance of Theorem 3 is illustrated.)

Because expert 4's expected unit compensation $y_4 = 5$ is equal to the value of the moderator's unit cost $c_4 = 5$, his/her adjusted optimal opinion $o_4^* = 9$ is no more than his/her original opinion $o_4 = 10$. Likewise, because expert 1 (expert 2)'s expected unit compensation $y_1 = -1$ ($y_2 = -4$) is equal to the opposite value of the moderator's unit cost $c_1 = 1$ ($c_2 = 4$), expert 1 (expert 2)'s adjusted optimal opinion $o_1^* = 9$ ($o_2^* = 6$) is no less than his/her original opinion $o_1 = 0$ ($o_2 = 3$). The absolute value of expert 3's expected unit compensation $|y_3| = 2$ is lower than the value of the moderator's unit cost $c_3 = 3$ and his/her adjusted optimal opinion $o_3^* = 6$ is equal to his/her original opinion $o_3 = 6$. (The economic significance of Theorem 4 is illustrated.)

The relationships between the directions of individual opinions adjustments and their compensations of Theorem 3 and Theorem 4 in the numerical example are shown in Fig. 6.

In model (17), because expert 2 (expert 3)'s adjusted optimal opinion equals to the minimum value ($o_{il} = 6$) of the acceptable consensus range, the part of expert 2 (expert 3)'s expected unit compensation due to the limitation of the maximum value of the consensus range is 0, that is $y_6 = 0$ ($y_7 = 0$). Likewise, expert 1 (expert 4)'s adjusted opinion equals to the maximum value ($o_{ir} = 9$) of the acceptable consensus range, the part of expert 1 (expert 4)'s expected unit compensation due to the limitation of the minimum value of the consensus range is 0, that is $y_9 = 0$ ($y_{12} = 0$). (The economic significance of Theorem 6 is illustrated.)

In model (17), because expert 1 (expert 4)'s expected partial unit compensation is $y_5 = -0.5$ ($y_8 = -6$) due to the limitation of the maximum value of the consensus interval, his/her adjusted opinion must be equal to the maximum value of the acceptable range of opinions, which is $o_1^* = 9$ ($o_4^* = 9$). Likewise, because expert 2's expected partial unit compensation is $y_{10} = 3.5$ due to the limitation of the minimum value of the consensus interval, his/her adjusted opinion must be equal to the minimum value of the acceptable range of opinions, which is $o_2^* = 6$. Expert 3's expected first part of unit compensation is 0, his/her adjusted opinion is located in the acceptable range of opinions, that is $o_3^* = 6 \in [6, 9]$. (The economic significance of Theorem 7 is illustrated.)

5. Application in loan consensus of online P2P lending

5.1. Problem description

P2P lending is one of the new forms of Internet financial services. P2P lending websites provide platforms for borrowers and lenders. Borrowers submit loan applications to P2P platforms, and lenders select loan applications that meet their expectations. When a borrower completes a loan, the platform puts lenders in contact with borrowers by charging a fee. P2P platforms charge service fees for transactions, rather than setting higher interest rates than the cost of money as traditional financial institutions.

By reducing the middle man, P2P lending is beneficial for both borrowers and lenders. Borrowers can conveniently get loans at lower interest rates, and lenders can get higher returns for their money. One of features of P2P lending is that a lender needs to allocate money to different borrowers to reduce risk and maximize returns, which is difficult because many of them have no expertise in finance (Guo, Zhou, Luo, Liu, & Xiong, 2016; Serrano-Cinca & Gutiérrez-Nieto, 2016; Ahmed et al., 2018; Asongu et al., 2018). On the other hand, borrowers are often unable to find appropriate lenders and lead to loss of biddings, which cause potential losses to platforms. In Zhang et al. (2018), the authors studied a famous U.S. P2P lending platform, Prosper, and found that more than 60% of failed loans received less than 20% of their funding goals. In Liu, Wen, and Qin (2017), the authors researched on one of the largest Chinese P2P lending platforms, (12:italic)PPDai.com/(12:italic), and found that only 20% of all bids on the platform can achieve full bids, and a large number of loan requests cannot be granted. How to coordinate the interests of lenders and borrowers to reach a consensus is an important task in P2P lending.

This P2P lending scenario can be formulated as a minimum cost soft consensus problem. We can establish a minimum cost model and its maximum return model under a certain degree of consensus. This application treats a lender and a certain number of borrowers as experts. Lenders expect higher returns by lending money to borrowers than to banks, and borrowers want to pay lower costs to get loans from lenders than from banks. Though lenders and borrowers have different expectations about interest rates, they are potentially able to reach a consensus about loans. A P2P platform

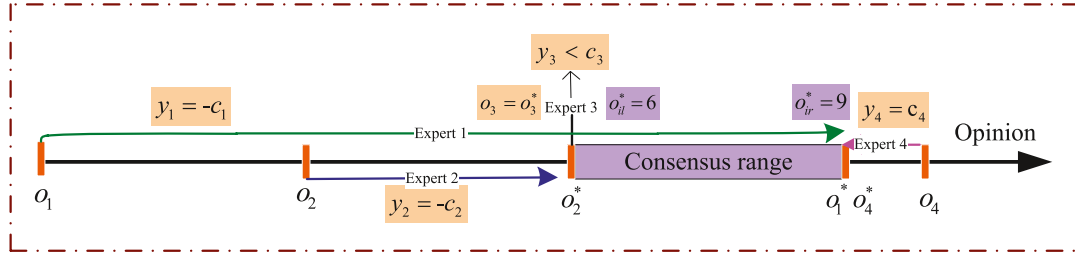


Fig. 6. The economic significances of Theorem 3 and Theorem 4 in the numerical example.

is regarded as a moderator, who develops compensation strategies to make borrowers change their expected interest rates, and helps lenders and borrowers reach a certain level of consensus.

Many Chinese P2P lending platforms, such as [PPDai.com](#), [Jiedaibao.com](#), and [Renrendai.com](#), charge borrowers relatively high service fees and allowance for risk reserves (average about 28% of the total loan). Compensations to borrowers can be regarded as platforms' rebates or discounts for these fees. Compensation strategies are used by platforms when a lender cannot find a suitable borrower under automatic bids or recommended systems.

The loan consensus in P2P lending platform can be described as follows. Suppose that there are m loan applications $e_1, e_2, \dots, e_m$ in a P2P lending website. The total amounts of the loans are W . Borrowers' expected interest rates are expressed as $o_i, i \in M = \{1, 2, \dots, m\}$, and the adjusted interest rates for the borrowers are denoted as $o'_i, i \in M = \{1, 2, \dots, m\}$. When a borrower changes the unit interest rate by 1%, the compensation provided by a P2P platform to the borrower is c_i . Let $w_i, i \in M = \{1, 2, \dots, m\}$ denote the weights of different amounts of loans, which satisfy $\sum_{i=1}^m w_i = 1$. Aggregation operator is used to aggregate different borrowers' opinions of interest rates and obtain a collective opinion o' , which is used to measure a lender's expected rate of return. The acceptable consensus level is γ , which indicates that a lender's desired interest rate is flexible. When $\gamma = 1$, it means that a loan consensus can be reached only when a lender and a borrower have the same expected interest rate. When $\gamma = 0$, a lender accepts any bid. In reality, the acceptable consensus level is between 0 and 1, which is the situation discussed in this article.

5.2. Example of the soft cost consensus models for a real-world P2P lending platform

Using data from [PPDai.com](#), a Chinese P2P platform, as an example, this subsection shows the construction of consensus cost models under a certain consensus degree. Suppose a lender has a loan of $M=10,000$ RMB. Four loans with different credit ratings (AA, A, B, and C, respectively) are selected as borrowers. The four borrowers' expected interest rates are: $o_1 = 11, o_2 = 16, o_3 = 22, o_4 = 28$ (Unit: %), the data of interest rates are from [12:italic PPDai.com/12:italic](#). Assuming that the platform's compensation strategy for four different credit levels is that when a borrower's interest rate is adjusted by 1%, the platform gives compensations of $c_1 = 1.5, c_2 = 1.0, c_3 = 2.5$, and $c_4 = 2.0$ (Unit: 10RMB). According to the default risk of a loan, the lender's allocation of funds is 4000, 3000, 2000, and 1000 (Unit: RMB) to the four loan applications, respectively. The lender's consensus level is $\gamma = 0.8$. Because the borrowers' opinions of interest rates present an additive relationship and the mean of them is appropriate to measure the lender's expected rate of return, AWA0 is used to aggregate borrowers' opinions and obtain a collective opinion. We adopt the MCC-AWA0- γ model in this

example. The soft MCC model for the loan consensus of online P2P lending can be constructed as follows:

$$LPM_2 : \min \phi = 1.5u_1 + 1.5v_1 + u_2 + v_2 + 2.5u_3 + 2.5v_3 + 2u_4 + 2v_4$$

$$\begin{cases} o'_1 - u_1 + v_1 = 11 \\ o'_2 - u_2 + v_2 = 16 \\ o'_3 - u_3 + v_3 = 22 \\ o'_4 - u_4 + v_4 = 28 \end{cases} \quad (18-1)$$

$$\text{s.t.} \begin{cases} 1 - \frac{|o'_1 - o'_2|}{\sigma'} \geq 0.8 \\ 1 - \frac{|o'_2 - o'_3|}{\sigma'} \geq 0.8 \\ 1 - \frac{|o'_3 - o'_4|}{\sigma'} \geq 0.8 \\ 1 - \frac{|o'_4 - o'_1|}{\sigma'} \geq 0.8 \end{cases} \quad (18-2)$$

$$0.4o'_1 + 0.3o'_2 + 0.2o'_3 + 0.1o'_4 - o' = 0 \quad (18-3)$$

$$o'_i, o', u_i, v_i \geq 0, i = 1, 2, 3, 4 \quad (18-4)$$

$$(18)$$

The unique solution to Model (18) is $\bar{X}^* = (3.67 \ 0 \ 3.56 \ 0 \ 0 \ 0 \ 0 \ 6 \ 14.67 \ 19.56 \ 22 \ 22 \ 18.33)^T$, and the optimal value of the objective function of Model (18) is $\min \phi = 21.1$. The optimal interest rates for the four borrowers are $o'_1 = 14.67\%, o'_2 = 19.56\%, o'_3 = 22\%, o'_4 = 22\%$. The optimal interest rate for the lender is $o' = 18.33\%$, which satisfies the expected loan consensus level $\gamma = 0.8$. The compensation paid by the platform to reach the loan consensus is 211 RMB. It is obvious that the lender's expected interest rate increased by 1.93%, and the profit increased by 11.79%. To coordinate the transaction of 10000RMB, the subsidy fee paid by the platform is only 211RMB, which accounts for 2.1% of the transaction turnover. The result of the soft MCC model for the loan consensus of online P2P lending under different consensus levels is summarized in Table 4.

The dual problem of Model (18) is:

$$DLPM_2 : \max \psi = 11y_1 + 16y_2 + 22y_3 + 28y_4$$

$$\begin{cases} |y_1| \leq 1.5 \\ |y_2| \leq 1.0 \\ |y_3| \leq 2.5 \\ |y_4| \leq 2.0 \end{cases} \quad (19-1)$$

$$\text{s.t.} \begin{cases} y_1 + y_5 + y_9 + 0.4x \leq 0 \\ y_2 + y_6 + y_{10} + 0.3x \leq 0 \\ y_3 + y_7 + y_{11} + 0.2x \leq 0 \\ y_4 + y_8 + y_{12} + 0.1x \leq 0 \end{cases} \quad (19-2)$$

$$-x - 1.2y_5 - 1.2y_6 - 1.2y_7 - 1.2y_8 \quad (19-3)$$

$$-0.8y_9 - 0.8y_{10} - 0.8y_{11} - 0.8y_{12} \leq 0 \quad (19-4)$$

$$y_5, y_6, y_7, y_8 \leq 0, y_9, y_{10}, y_{11}, y_{12} \geq 0 \quad (19-5)$$

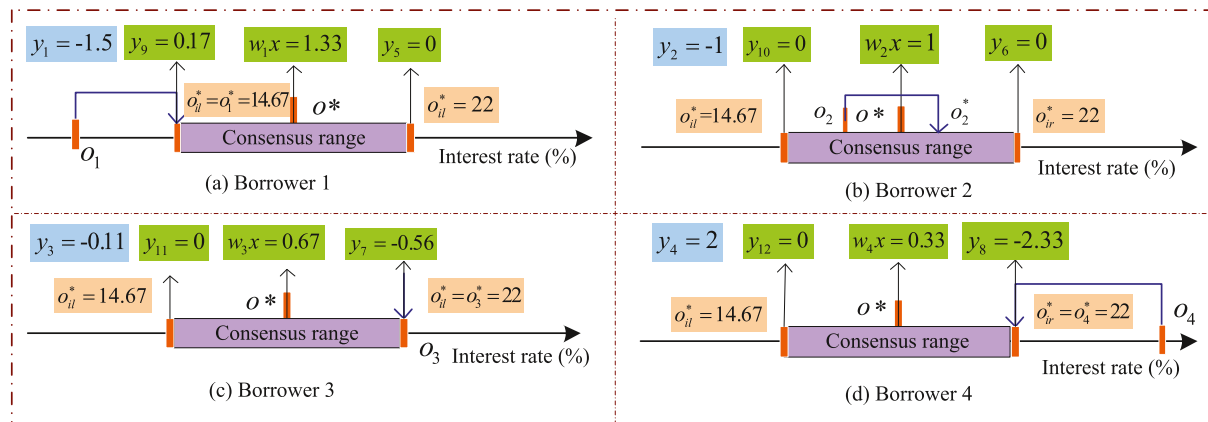
$$(19)$$

Model (19) has the unique solution $\bar{Y}^* = (-1.5 \ -1 \ -0.11 \ 2 \ 0 \ 0 \ -0.56 \ -2.33 \ 0.17 \ 0 \ 0 \ 0 \ 3.33)^T$, and its maximum objective function is $\max \psi = 21.1$. That is, the unit compensations expected by the

Table 4

The result of the soft MCC model for the loan consensus under different consensus levels.

	$\gamma = 0.1$	$\gamma = 0.2$	$\gamma = 0.3$	$\gamma = 0.4$	$\gamma = 0.5$	$\gamma = 0.6$	$\gamma = 0.7$	$\gamma = 0.8$	$\gamma = 0.9$	$\gamma = 1.0$
o_1^*	11	11	11	11	11	11	11.85	14.67	18	22
o_2^*	16	16	16.24	19.67	23.56	16	18.62	19.56	20.67	22
o_3^*	22	22	22	22	22	22	22	22	22	22
o_4^*	28	28	28	28	28	22.14	22	22	22	22
o^*	16.40	16.40	16.47	17.50	18.67	15.81	16.92	18.33	20	22
ϕ	0	0	0.2	3.7	7.6	11.7	15.9	21.1	27.2	34.5

**Fig. 7.** Relationships between each borrower's interest rate adjustment and expected unit compensation.

four borrowers are $y_1 = -1.5$, $y_2 = -1$, $y_3 = -0.11$, and $y_4 = 2$. The positive or negative value represents the interest rate direction that borrowers should adjust to achieve the loan consensus. The total compensation expected by the four borrowers is 211 RMB. It indicates that the total compensation expected by the borrowers is equal to the cost paid by the platform in model (18) and the soft loan consensus is reached.

Section 4.4 describes the relationships between the soft minimum cost consensus model and maximum return model using a numerical example. The following section further explains the relationships between each expert's opinion adjustment and expected unit compensation by using the loan consensus example above. For borrower 1, the value of unit expectation compensation is negative, indicating that borrower 1 adjust his/her interest rate in the increased direction to achieve the loan consensus. The absolute value of the unit compensation expected by borrower 1 is equal to the unit payment provided by the P2P lending platform ($|y_1| = c_1 = 1.5$ (10 RMB)), indicating that the soft consensus for compensation has been reached.

In addition, the unit compensation of borrower 1 consists of two parts. One part is 0.17 (10RMB), due to the limitation of the minimum value of the loan consensus interval. That is, to satisfy the established loan consensus level 0.8, borrower 1 needs to increase his/her expected interest rates, and therefore needs to be compensated. The unit compensation is 0.17 (10 RMB). The other part comes from the fact that borrower 1 makes contribution to form the optimal loan consensus opinion (second part of compensation) and the unit compensation is 1.33 (10 RMB). In other words, borrower 1 modified his/her interest not only to reach the loan consensus interval, but also to satisfy the specific relationship with the group interest rate in order to reach the loan consensus. Borrowers should be compensated because they contribute to the composition of the group consensus. The second part of compensation is obviously presented by borrower 2, whose original interest rate is in the value of the loan consensus interval, and the opinion is modified only because the second part of compensation. The compensation mechanisms for borrow 2, 3, and 4 can be illustrated

similarly. The opinion adjustment and the corresponding unit composition for the four borrowers are shown in Fig. 7.

Although our study is based on a Chinese P2P lending platform, it is applicable to other lending platforms. This method can be used by the platforms when a lender cannot find a suitable borrower under automatic bids. It can help platforms to increase the efficiency in matching the supply and demand of funds in P2P lending.

6. Conclusions

Reaching a unanimous consensus is often costly and unnecessary in real life GDM situations. Based on this observation, this paper proposed generalized soft cost consensus models under a certain degree of consensus by defining a consensus level function and a generalized aggregation operator. We built three soft minimum cost consensus models to show the changes of consensus costs and individual expert's opinions under different consensus levels with the linear and non-linear models. Furthermore, using dual linear programming theories, we constructed the corresponding maximum return model under a certain degree of consensus. The relationships between the soft minimum cost and maximum return models and the economic significance of the models were also discussed. To validate its usefulness in real-world GDM problems, we applied the proposed soft consensus models to loan consensus problems in P2P lending using data from a Chinese P2P platform. It showed that the proposed models can assist lenders and borrowers to reach a certain level of consensus about interest rates, and increase the efficiency in matching the supply and demand of funds in P2P lending.

This article made three contributions. First, it proposed the concepts of soft minimum cost consensus, which extended the minimum cost consensus theory proposed by Ben-Arieh and Easton (2007) and Ben-Arieh et al. (2009) to a certain consensus level, improved the minimum cost and maximum return models (Gong, Xu, Zhang, et al., 2015; Gong, Zhang et al., 2015), and explained the economic meaning of soft consensus models. Sec-

ond, this paper also provided a way to show the consumption of resources and the changes of experts' opinions under various consensus degrees by constructing three basic soft minimum cost consensus models, which are useful for optimizing resource allocations. Third, this paper studied the relationships among experts' expected unit compensation, compositions of an expert's expected unit compensation, and his/her adjusted opinions guided by a moderator. It contributes to the study of compensation mechanism among a moderator and experts under a certain degree of consensus, and is beneficial for reaching a soft consensus.

In some consensus decision-making problems, experts' opinions are uncertain. To describe this uncertainty, fuzzy information is widely used to express the preferences of experts (Cabrerizo et al., 2010; Gong, Xu, Zhang, et al., 2015; (Liu et al., 2012; Wan, Wang, & Dong, 2017), especially when crisp value cannot completely represent the real viewpoints of experts. One of our future research directions is to investigate the characteristics of consensus when interval preferences are used. The determination of a reasonable unit compensation cost for experts in the minimum cost consensus problem is another research topic, and a few researches have done some work in this area Cheng et al. (2018). In soft consensus cost problems, how to formulate unit compensation prices for experts requires further study.

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